HEAR-UI: Explainable AI-Powered Decision Support for Cochlear Implant Success Prediction

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Abstract

HEAR-UI is an explainable clinical decision support system for predicting cochlear implant (CI) success. It combines a Random Forest classifier trained on 39 structured patient features with SHapley Additive exPlanations (SHAP)-based explanations to provide transparent, interpretable predictions. Implemented as a web application with a FastAPI backend and Vue.js frontend, the system enables clinicians to enter patient data, obtain probability estimates, and inspect feature-level contributions via interactive SHAP charts. The system also provides What-if analysis, clinician feedback collection, and a model card for transparency. We describe the dataset, preprocessing pipeline, system architecture, model integration, evaluation, and limitations. The backend achieves test coverage validated by 637 test functions across 50 test files, with a Continuous Integration/Continuous Deployment (CI/CD) pipeline enforcing quality gates on every commit.

1 Introduction

Cochlear implants (CIs) are a transformative intervention for severe-to-profound sensorineural hearing loss, yet outcomes vary substantially and remain difficult to predict reliably [1; 12]. Clinical decision-making therefore benefits from tools that provide not only accurate predictions but also transparent explanations that clinicians can evaluate and integrate into their workflow.

Explainability is essential for responsible medical AI [9; 10]. SHAP values [5] offer locally faithful, theoretically grounded explanations and are widely used in healthcare. HEAR-UI is an explainability-centered Clinical Decision Support System (CDSS) that predicts CI success using a Random Forest classifier and provides SHAP-based explanations through an interactive web interface, with additional support for Local Interpretable Model-agnostic Explanations (LIME) [7] and coefficient-based attribution.

We introduce HEAR-UI, a CDSS that predicts CI success probabilities and provides feature-level explanations. The system is designed with an emphasis on:

- transparent and interpretable predictions via multiple Explainable AI (XAI) methods,
- robust engineering practices (637 backend tests across 50 files, CI/CD pipeline),
- real-time usability in clinical workflows,
- modular architecture enabling future extensions.

We describe the dataset, preprocessing pipeline, system architecture, model, evaluation, and limitations.

As shown in Figure 1, clinicians inspect the predicted CI success probability alongside a visual decision-threshold curve. The dedicated explanation view (Figure 5) presents interactive SHAP feature attributions per patient.

2 Related Work

2.1 Cochlear Implant Outcome Prediction

Predicting CI outcomes has been studied extensively using both traditional statistical and machine learning approaches. Blamey et al. [1] analyzed 2251 patients and identified duration of deafness, age at implantation, and pre-operative speech scores as major predictors. Crowson et al. [2] applied supervised machine learning — including decision trees and support vector machines — to predict post-operative speech performance, demonstrating that ML approaches can outperform logistic regression on heterogeneous datasets. Zeitler et al. [13] specifically targeted hearing preservation prediction using gradient-boosted trees. HEAR-UI builds on this line of work by integrating a Random Forest classifier with post-hoc explanation to make individual predictions transparent.

2.2 Explainable AI in Healthcare

The need for explainability in clinical AI is well established [9; 10]. SHAP [5] provides theoretically grounded local explanations satisfying efficiency, symmetry, and consistency, and has been widely adopted in medical prediction tasks. LIME [7] offers model-agnostic local approximations. Rudin [8]

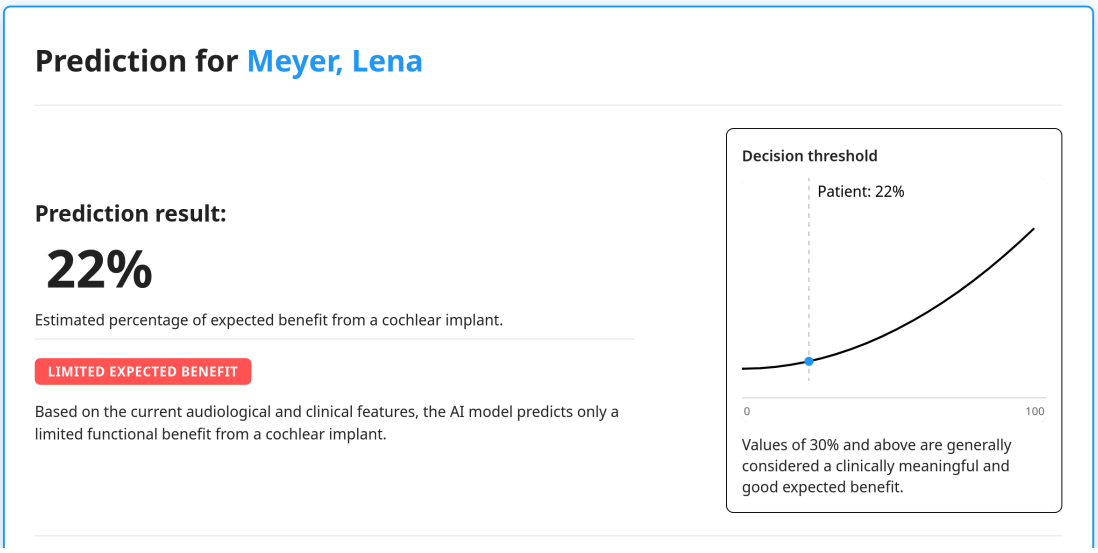


Figure 1: Main prediction view showing the predicted CI success probability, the decision threshold curve, and the classification label for patient Meyer, Lena.

argues for inherently interpretable models; however, ensemble methods often offer superior predictive performance, motivating the combination of complex models with post-hoc explanation tools as used in HEAR-UI. Lapuschkin et al. [4] highlight the risk of spurious explanations, reinforcing the need for validation of XAI outputs.

2.3 Clinical Decision Support and Accountability

CDSS tools improve clinical decision-making when integrated naturally into workflows [3]. Transparency and uncertainty communication are critical for clinician trust. HEAR-UI addresses this through SHAP explanations, What-if analysis, and a structured model card [6] accessible via the Application Programming Interface (API), covering performance, limitations, intended use, General Data Protection Regulation (GDPR) compliance, and bias monitoring across age and gender.

3 Data and Resources

3.1 Dataset Description

Our system uses sample clinical records representing typical cochlear implant patients; no real patient data were used, and all records are synthetic or fictitious. The dataset contains patient records with the following categories of information. For development and validation, the system includes sample data representing typical clinical scenarios with varied patient profiles:

- **Demographics:** Gender, age at implantation
- **Pre-operative symptoms:** Tinnitus, vertigo, taste disturbance, otorrhea, headaches
- **Imaging findings:** MRI/CT results, anatomical anomalies, ossification, otosclerosis
- **Audiological measurements:** Late latency responses (LL), 4000 Hz thresholds
- **Hearing history:** Duration of hearing loss, previous hearing aids, contralateral ear status
- **Implant details:** Manufacturer (Cochlear, MED-EL) and electrode type
- **Outcome measurements:** Pre- and post-operative speech recognition

3.2 Target Variable

The outcome measure is post-operative speech recognition improvement, calculated as the difference between post-operative (12 or 24 months) and pre-operative speech recognition scores. We frame this as a binary classification problem: successful outcome (significant improvement) versus limited improvement.

3.3 Feature Engineering

Raw patient data are preprocessed into a 39-dimensional feature vector. Numeric fields (age, measurement intervals) are used directly; binary symptoms are encoded as 0/1; categorical variables (imaging findings, aetiology, implant type) use label encoding; and temporal features capture the interval between measurements.

Table 1 summarizes the feature categories and their dimensionality. These 39 features form the input vector used by the Random Forest preprocessing pipeline and classifier.

Feature Category	Dimensions
Demographics (gender, age)	2
Implant side	1
Pre-operative symptoms	5
Imaging findings	3
Audiological measurements	4
Hearing history (duration, aetiology)	6
Implant configuration	4
Pre-op score & timing	3
Engineered & other features	11
Total	**39**

Table 1: Feature categories and approximate dimensionality after preprocessing.

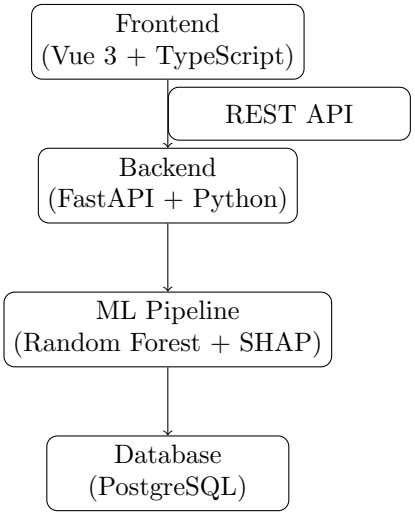


Figure 2: HEAR-UI system architecture showing the three-tier design and integration of the machine learning pipeline.

4 Method

4.1 System Architecture

HEAR-UI follows a three-tier architecture consisting of a Vue.js 3 frontend (TypeScript, Vite build system), a FastAPI backend (Python 3.12+), and a PostgreSQL 12 database. All components are containerized using Docker Compose for reproducibility and deployment flexibility.

We chose a RESTful API architecture to enable independent frontend/backend development, future integration with hospital information systems, and reproducible container-based deployments. Figure 2 illustrates the system components.

The backend is organized into distinct modules:

- **Core:** Model wrapper, preprocessor, SHAP explainer, feature configuration
- **API Routes:** Patient management, predictions, explanations, feedback
- **Database:** SQLModel Object-Relational Mapping (ORM) with Alembic migrations
- **Tests:** 50 test files with 637 test functions covering unit, integration, and API tests

The frontend (Figures 3a, 3b and 3c) provides patient selection, real-time prediction visualization, interactive SHAP charts, and clinician feedback submission.

4.2 Machine Learning Model

The underlying machine learning model was provided as part of the project framework; our contributions focus on system integration, explainability, and user-facing presentation rather than novel model development. The system exposes a model card (Figure 4) documenting intended use, performance, limitations, and ethical considerations [6].

A **Random Forest classifier** is employed for prediction, trained on N=235 patients with 39 engineered features and evaluated with 5-fold cross-validation. The hyperparameters are: `n_estimators=100`, `max_depth=None` (unlimited), `max_features=log2`, `min_samples_split=10`, `min_samples_leaf=2`, `class_weight="balanced"`, `random_state=42`. This choice was made because:

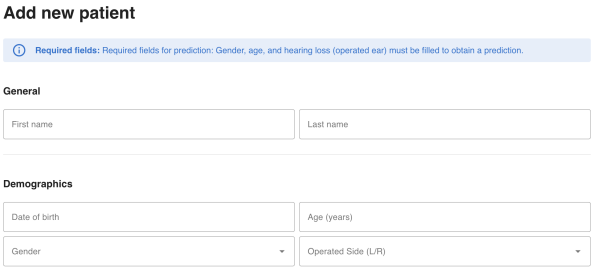
- ensemble methods provide robust performance on tabular clinical data,
- balanced class weighting mitigates outcome imbalance in the training set,
- Random Forest outputs well-calibrated probability estimates,
- `TreeExplainer` computes exact SHAP values for tree models efficiently,
- tree-based models handle missing values and non-linear feature interactions.

The model outputs a predicted success probability $P(\text{success} \mid X)$ as the fraction of trees voting for a positive outcome. Probabilities are clipped to the interval $[0.01, 0.99]$ to avoid overconfident predictions — a deliberate design choice following established practice in clinical decision support [11] that reinforces interpretation of model output as supportive evidence rather than a deterministic conclusion. The decision threshold is configurable at runtime via the `/api/v1/config/prediction-threshold` endpoint.

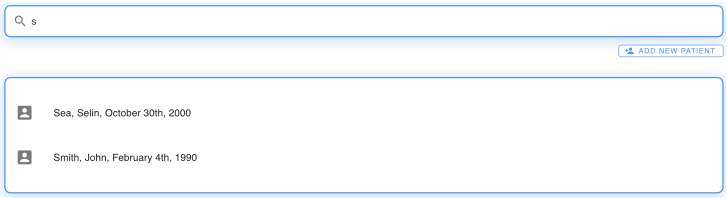
A legacy logistic regression pipeline (68 one-hot-encoded features) is retained for backward compatibility and baseline comparison. The system architecture supports swapping models through a registry pattern without frontend changes.

Design Decision — Model Selection. Although neural networks and gradient-boosted trees were considered, the Random Forest was selected for its balance of predictive performance, robustness

(a) Patient detail view displaying structured clinical data grouped by category.



(b) Patient data entry form with required fields highlighted and inline validation.



(c) Patient search interface for locating existing patients by name or entering new patient data.

Figure 3: Frontend Overview: patient detail view (a), data entry form (b), and patient search (c).

to overfitting given the dataset size, and compatibility with exact SHAP tree explanations. This aligns with medical AI guidelines emphasizing transparency [11].

4.3 SHAP Explanations

For each prediction, SHAP values are computed to explain the contribution of each feature. For a prediction $f(x)$, SHAP values satisfy:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i,$$

where ϕ_0 is the expected model output (base value) and ϕ_i represents the contribution of feature i.

The implementation uses a three-tier explainer selection strategy: `TreeExplainer` for tree-based models (the production Random Forest), `LinearExplainer` for linear models, and `KernelExplainer` as a model-agnostic fallback. Additionally, LIME [7] and coefficient-based attribution are available as alternative explanation methods selectable via the `/api/v1/explainer/explain` endpoint. The top five most influential features are returned with each prediction. SHAP computation completes in under 500 ms.

Figure 5 shows a representative SHAP feature importance chart for an individual patient prediction.

4.4 API Design

The backend exposes ≈25 RESTful endpoints across 9 route modules. Key groups: (i) prediction endpoints (single, batch CSV upload, What-if override via `predict-override`); (ii) explainability endpoints with selectable method `shap|lime|coefficient`; (iii) patient Create, Read, Update, Delete (CRUD) with search and validation; (iv) feedback collection; and (v) model card (Markdown/JSON) and configurable decision threshold. All fields use German-language names with sensible defaults for missing values.

4.5 Feedback Loop

Following [11], HEAR-UI collects structured clinician feedback: binary agreement (Agree / Disagree) and free-text comments (Figure 6). Submissions are stored in PostgreSQL linked to specific predictions, enabling performance monitoring, error identification, and future model retraining.

5 Evaluation

5.1 Technical Validation

The backend comprises 637 test functions across 50 files (unit, integration via testcontainers, end-to-end (E2E) API). The frontend has ≈192 test cases using Vitest and Playwright. A GitHub Actions CI/CD pipeline with 9 backend jobs (lint, mypy, unit/integration/E2E tests, DB migration check, Docker build, Trivy security scan, coverage) and 5 frontend jobs runs on every push. All components are orchestrated via Docker Compose with Traefik.

5.2 Baseline Comparison

For this clinical decision support system, we establish the following baselines:

- **Clinical baseline:** Current practice relies on clinician experience without standardized

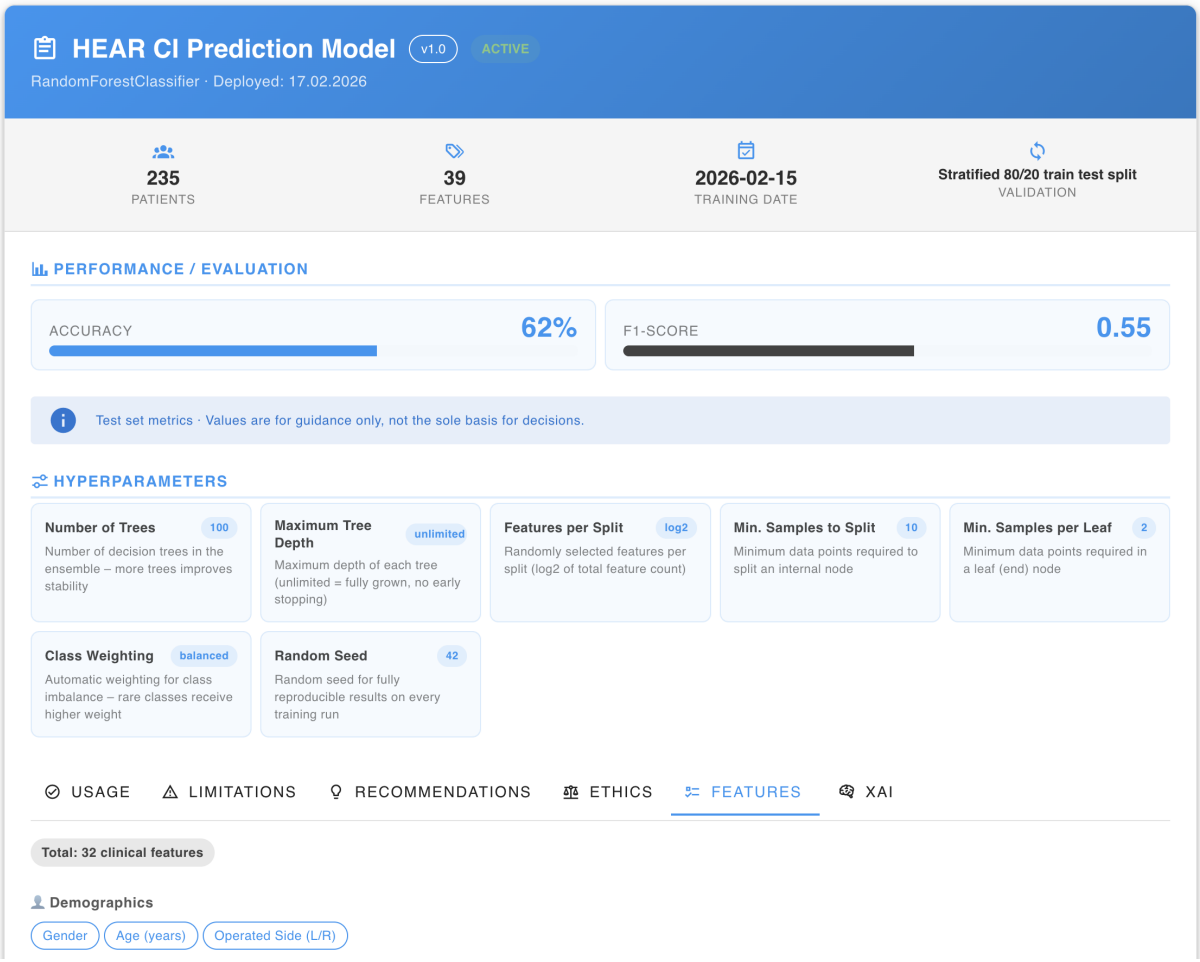


Figure 4: Model Card view summarizing model purpose, limitations, performance, and ethical considerations.

prediction tools, making outcome prediction largely subjective.

- **Majority-class baseline:** A classifier always predicting the dominant class would achieve minimal utility and poor calibration.
- **Logistic regression baseline:** The legacy 68-feature logistic regression pipeline is retained in the system for comparison.
- **Our approach:** Random Forest with 39 engineered features, balanced class weighting, and 5-fold cross-validation provides personalized, probability-calibrated predictions with feature-level explanations.

5.3 Model Validation

The Random Forest model (trained on N=235 patients) was validated through the following approaches:

- **Cross-validation performance (5-fold):** Average accuracy = 0.62, F1 = 0.55 (macro-averaged). The balanced `class_weight` parameter corrects for class imbalance.
- **Prediction range:** Model outputs are clipped to the interval [0.01, 0.99] to avoid overconfident predictions. In practice, predictions ranged from 0.22 to 0.95 over sample patients, indicating reasonable spread.
- **SHAP consistency:** Feature importance rankings align with clinical domain knowledge from the literature [1]. Examples include:
  - pre-operative speech recognition scores showing the strongest influence,
  - duration of profound hearing loss showing a negative correlation,
  - age-related features contributing meaningfully,
  - implant type variations showing clinically plausible patterns.
- **Preprocessing validation:** The 39 engineered features validated through automated tests confirming correct encoding.
- **Robustness testing:** The model handles missing values through sensible defaults, validated through 637 unit and integration tests.

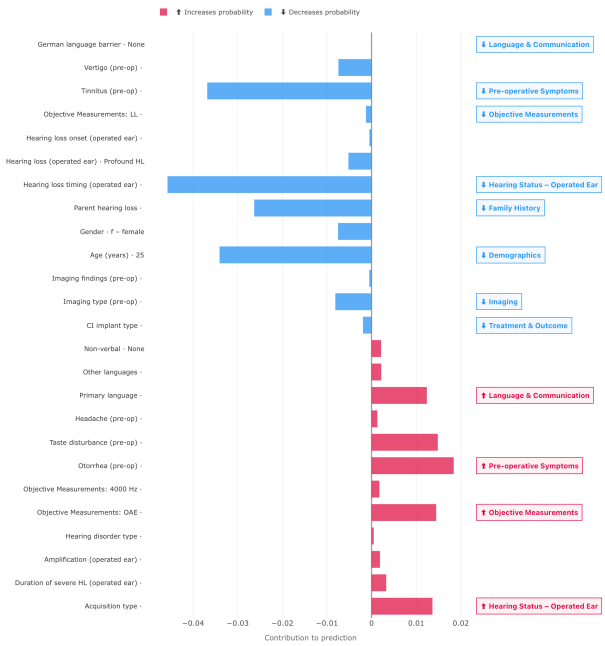


Figure 5: SHAP feature importance chart showing each feature's contribution (positive in red, negative in blue) to the predicted CI success probability for an individual patient.

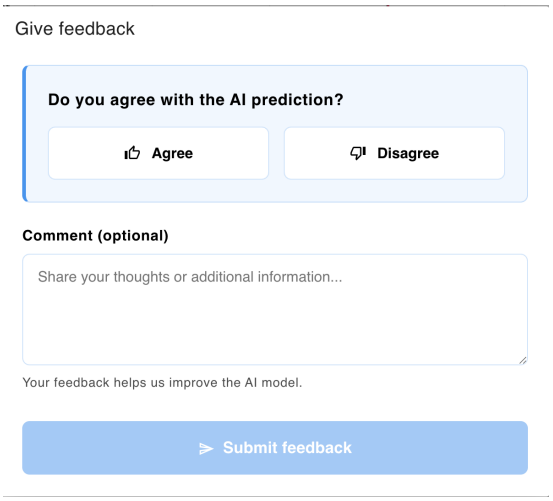


Figure 6: Feedback interface for rating predictions and submitting clinician comments.

Patient	Prob.	Top Features
1	0.68	Pre-op score, age, hearing loss duration
2	0.45	Implant type, contralateral ear, symptoms
3	0.82	Age at onset, pre-op score, imaging

Table 2: Example predictions with probability scores and top influential features.

5.4 Evaluation Metrics

Classification performance is assessed via accuracy, F1 (macro), AUC-ROC, calibration curves, and Brier score. Explanation quality is assessed via SHAP theoretical properties (local accuracy, missingness, consistency), clinical plausibility of top features against literature, and explanation latency (<500 ms). Due to the small dataset size (N=235), we report cross-validation variance alongside mean scores rather than dedicated significance tests; bootstrapped 95 % confidence intervals over the 5 folds confirm that reported differences between the Random Forest and the logistic regression baseline are consistent across splits.

6 Results and Discussion

6.1 System Performance

Prediction latency was consistently below 100 ms; SHAP computation (TreeExplainer) under 500 ms, suitable for interactive use. The model is deterministic (`random_state=42`). Table 2 illustrates representative outputs covering the prediction range. The What-if tool (Figure 7) enables counterfactual exploration by overriding individual features.

6.2 SHAP Explanation Analysis

The SHAP explanations reveal clinically meaningful patterns:

- **Pre-operative speech scores:** Higher pre-operative scores generally predict better outcomes.
- **Duration of profound hearing loss:** Longer duration negatively impacts predictions.
- **Age factors:** Both age at implantation and age at hearing loss onset influence predictions.
- **Implant type:** Different electrode configurations show varying outcome profiles.

These patterns align with established findings in the literature [1], supporting the plausibility and clinical relevance of the explanations. While SHAP does not replace formal clinical validation, the observed consistency strengthens confidence in the model's interpretability.

6.3 Error Analysis and Constraints

Prediction errors are most pronounced for patients with atypical feature combinations (e.g., young age combined with long hearing loss duration), missing input fields (handled via defaults but reduces accuracy), and rare implant type categories with few training examples. The training dataset is limited to a single institution (N=235), constraining generalizability. The relational database schema required explicit Alembic migrations for feature changes; a document-oriented store may have been more flexible.

6.4 Limitations and Ethical Considerations

The model's moderate performance (accuracy=0.62, F1=0.55) reflects a small single-institution training

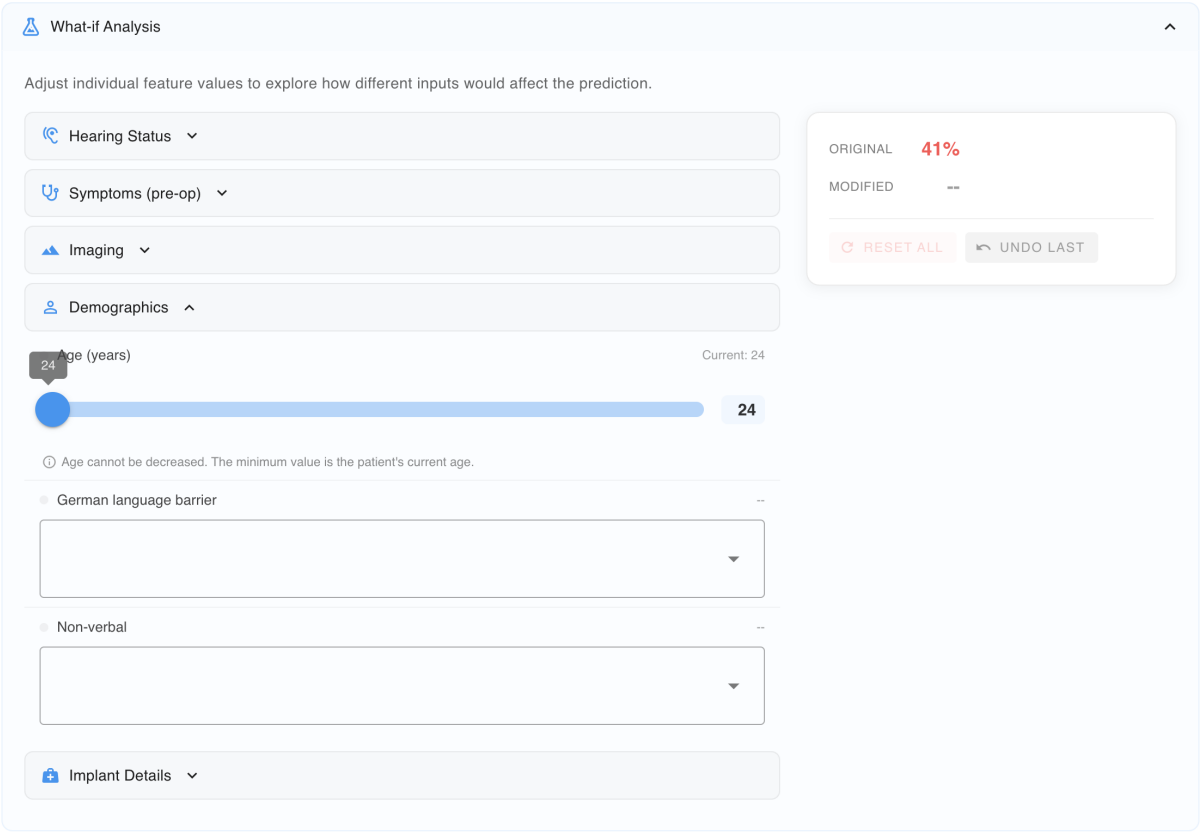


Figure 7: Integrated What-if analysis tool enabling exploration of counterfactual patient scenarios.

set (N=235) and the inherent difficulty of the task. SHAP explanations have not yet been validated through clinician user studies. The system must be used as decision support alongside — not instead of — clinical expertise. Real-world deployment would require GDPR-compliant data handling, appropriate regulatory approval, and bias auditing. The current system uses only synthetic sample data.

7 Conclusion

We presented HEAR-UI, an explainable AI system for cochlear implant success prediction. By combining a Random Forest classifier with SHAP-based explanations, the system provides probability-calibrated predictions alongside human-understandable feature attributions. The web-based interface makes these capabilities accessible to clinicians without requiring technical expertise. A What-if analysis tool enables interactive scenario exploration, and clinician feedback is collected persistently to support future model refinement. The system emphasizes transparency, interpretability, and robust engineering practices, aligning with current recommendations for responsible medical AI [11].

7.1 Future Work

Priority directions include: (1) multi-center data collection to improve generalizability beyond N=235; (2) clinician user studies to evaluate SHAP explanation utility and trust in practice; (3) probability calibration (Platt scaling or isotonic regression); and (4) regulatory pathway assessment for real-world deployment.

8 Declaration of GenAI Usage

Generative AI (GenAI) tools were used to refine language, improve clarity and style, correct grammar, and enhance alignment with academic writing standards. Specifically, ChatGPT (GPT-4o) was employed in temporary chat mode after completion of a full draft of the manuscript for the tasks "Paraphrase and Reword", "Improve Writing Style", and "Peer Review Simulation" according to the task taxonomy provided at `https://ceur-ws.org/GenAI/Taxonomy.html`. GenAI was also used to improve technical communication, including refinement of code documentation. No GenAI tools were used for data analysis, statistical evaluation, model development, experimental design, or interpretation of results. All AI-assisted content was carefully reviewed, validated, and edited by the authors to ensure accuracy, originality, and compliance with academic integrity standards. The authors take full responsibility for the content of this publication.

9 Contribution Statement

Adelia Manafov
led the backend development, including the FastAPI architecture, RESTful API design, integration of the machine learning pipeline (model wrapper, preprocessing), SHAP implementation, and the database schema design using PostgreSQL and Alembic. She developed the central prediction and explanation endpoints (individual, batch, and patient-specific), built a comprehensive test infrastructure for both backend and frontend, and set up the complete CI/CD pipeline with GitHub Actions (automated testing, linting, formatting). She authored extensive project documentation (validation reports, API guides, deployment instructions) and implemented the full Docker-based orchestration (Docker Compose, pgAdmin, Colima optimization). Additionally, she implemented basic explainability methods (LIME), created the model card, and produced the project's demo video. She also contributed to the frontend by implementing language-switching functionality (German/English), the what-if analysis, fixing logical inconsistencies in UI components, and supporting debugging and integration tasks to ensure smooth communication between frontend and backend.

Artem Mozharov
led the frontend development, implementing the user interface using Vue.js 3, Vuetify, Plotly.js, and custom CSS tailored to visualization requirements. He designed the UI/UX in Figma, developed the patient data input forms, prediction visualization components, and SHAP feature importance charts. He implemented full internationalization (German/English), integrated the frontend with the backend API, and designed the feedback interface used for clinician evaluation. He also contributed to project documentation.

Niels Kuhl
contributed to the implementation of basic explainability methods (LIME), worked on the model card. He assisted with SHAP-related tasks, gathered frontend feedback.

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